

UTOPIA ELECTRONIC DEVELOPMENT SPECIFICATIONS

Last updated 2/10/25

PROJECT OBJECTIVES

- Design a system of **ESP-32 microcontrollers** programmed to perform **waveform synthesis** and **audio effects** manipulated by **data** from **proximity sensors**.
- Integrate an **oscilloscope visual** for monitoring the performance of sound.
- Integrate **an external sound system** for the hardware synthesizer to output sound.

DEVELOPMENT FLOW

1. internal waveform generation to Speaker output
2. internal audio effect(s)
3. oscillator output
4. microphone input

COMPONENT SPECIFICATIONS

(x7) ESP32-WROVER-E

[datasheet](https://www.espressif.com/sites/default/files/documentation/esp32-wrover-e_esp32-wrover-ie_datasheet_en.pdf)

- 2 data processors, 3 audio processors, 1 oscilloscope processor, 1 audio output/master coordinator.
- 3.3V
- 8MB PSRAM, 8MB Flash

(x11) VL53L4CX Time-of-Flight Sensor

[listing](<https://www.adafruit.com/product/5425>)

- 3.3V
- I2C communication
- SCL, SDA, GPIO, XSHUT pins
- Range up to 13ft.

(x1) MAX9814 microphone

[documentation](<https://cdn-shop.adafruit.com/datasheets/MAX9814.pdf>)

(x2) MAX98357 Amplifier

[documentation](<https://learn.adafruit.com/adafruit-max98357-i2s-class-d-mono-amp>)

(x1) 128x64 STEMMA QT OLED display

[documentation](<https://learn.adafruit.com/monochrome-oled-breakouts>)

PCM1794A DAC

[datasheet](https://www.ti.com/lit/ds/symlink/pcm1794a.pdf?ts=1739053305998&ref_url=https%253A%252F%252Fhr.mouser.com%252F)

OPA1612 I/V CONVERTER

[datasheet](https://www.ti.com/lit/ds/symlink/opa1612.pdf?ts=1739154487554&ref_url=https%253A%252F%252Fwww.ti.com%252Fproduct%252FOPA1612)

- Converts DAC signal from current to voltage for the sub to read.

CUSTOM PCB

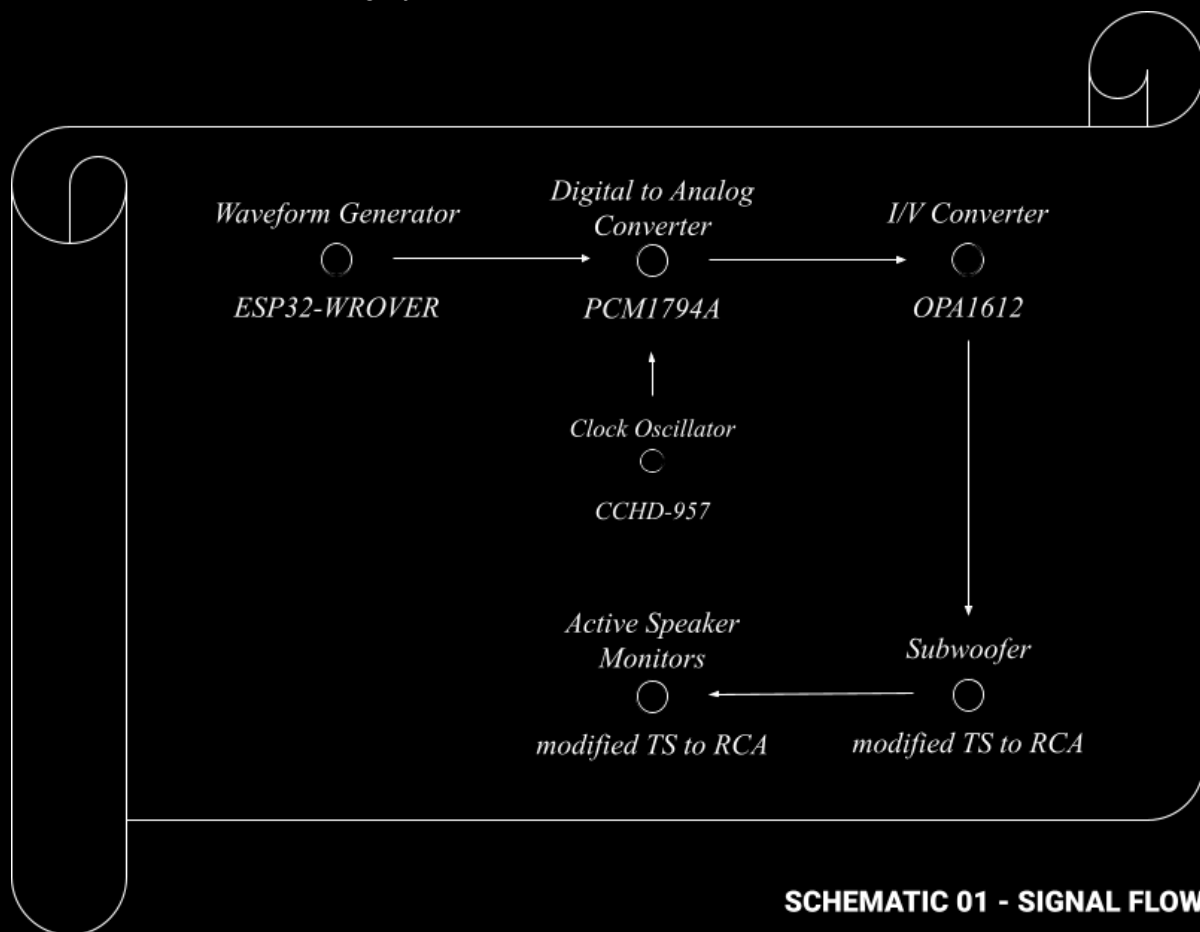
[resource](<https://jlcpcb.com/>)

CURRENT OBJECTIVES

01. SCHEMATIC: ESP32 waveform generator > DAC (Digital to Analog Converter) > I/V Converter > TS/RCA > Subwoofer > RCA/TS > monitor speakers

- Create custom symbols for:
 - DAC (PCM1794A)
 - I/V Converter (Burr Brown/OPA1612)
 - External Clock Oscillator (Crystek CCHD-957)

02. SCHEMATIC: Powering systems - Monitors, sub, ESP32s



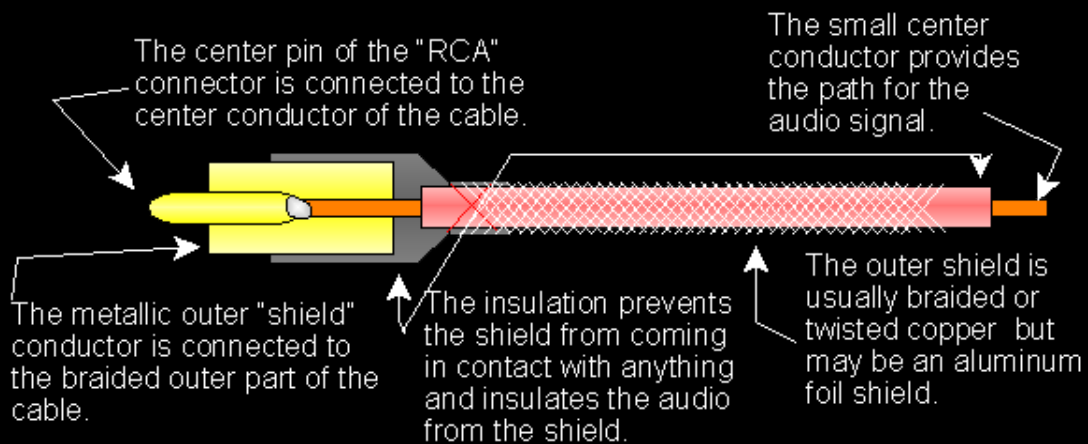
ESP32 WROVER > DAC PCM1794A > I/V Converter OPA 612 > Subwoofer > Active Monitor Speakers

Clock Oscillator Crystek CCHD-957 > DAC PCM1794A

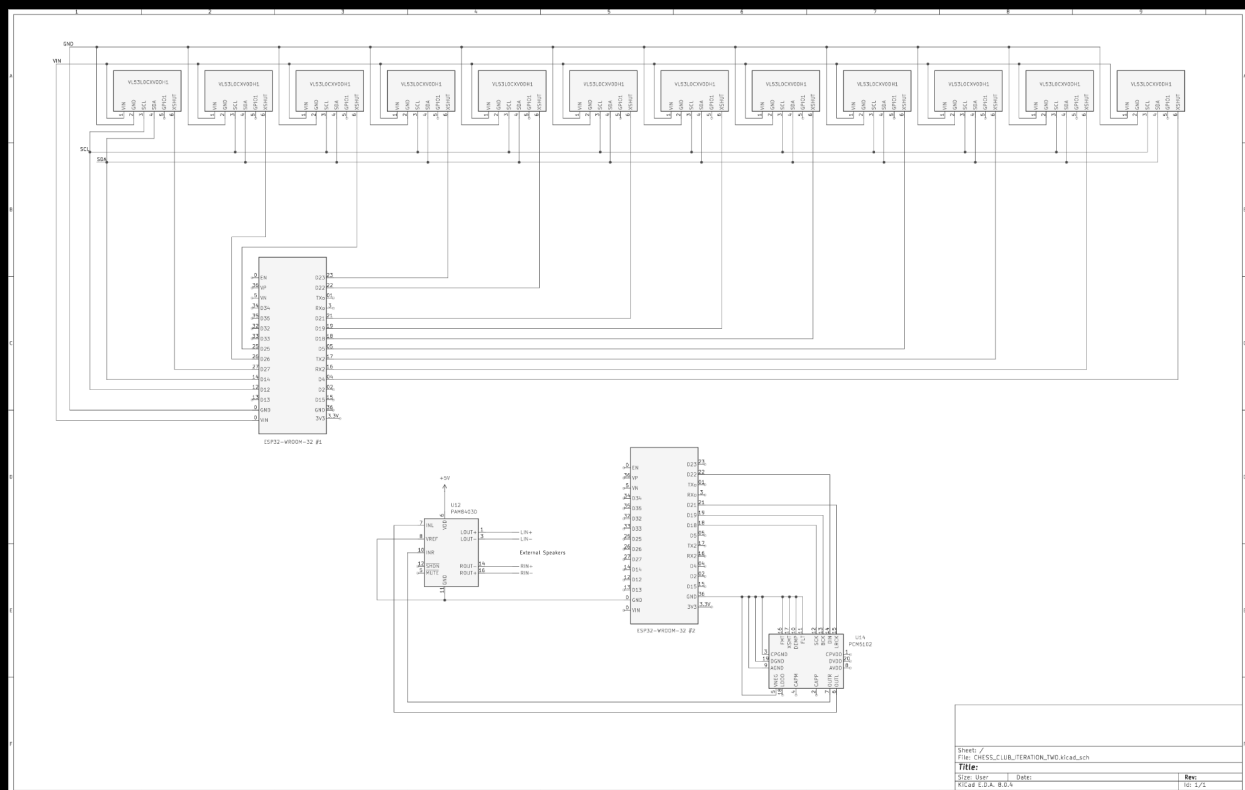
PROTOTYPE PINOUT

Notes: A transformer may be necessary in addition to shielded RCA cables

	OUT pin	IN pin	specification
lator)	GPIO 0 (I2S Data Out)	PCM1794A (DATA)	I2S data output from ESP waveform data.
	GPIO 25 (I2S Clock / BCK)	PCM1794A (BCK)	I2S bit clock (SCK) from
	GPIO 26 (I2S LRCLK)	PCM1794A (LRCK)	I2S Left/Right clock (LRCK)
	Pin 4 (VCC)	PCM1794A (Pin 13 or VCC)	Power supply for both DA
	Pin 2 (GND)	PCM1794A (Pin 3 or GND)	Ground connection.
	Pin 3 (OUT)	PCM1794A (SCK)	Master clock (MCLK) out
	Pin 7(SCK)	Crystek CCHD-957 (OUT)	Master clock (MCLK) from
	Pin 17 (R+ OUT)	OPA 612 #1 (IN+ A) Pin 3	Analog audio output to n
	Pin 18 (R- OUT)	OPA 612 #1 (IN- B) Pin 6	Analog audio output to th
	Pin 25 (L+ OUT)	OPA612 #2 (IN+ A) Pin 3	
	Pin 26 (L- OUT)	OPA1612 #2 (IN- B) Pin 6	
	Pin 1 (OUTA)	TS Tip Left to RCA	Positive line-level output
	Pin 1 (OUTA)	TS Tip Right to RCA	Positive line-level output
	Signal In + (Pos)	TS Tip Left to RCA	+ line-level signal input (F
	Signal In - (Neg)	TS Tip Right to RCA	- line-level signal input (R
	Signal Out + (Pos)	Monitor Speakers (Signal In +)	RCA output to active mo
	Signal Out - (Neg)	Monitor Speakers (Signal In -)	RCA output to active mo
	RCA Signal In + (Pos)	RCA Subwoofer (Signal Out +)	RCA input from subwoof
	RCA Signal In - (Neg)	RCA Subwoofer (Signal Out -)	RCA input from subwoof
<u>Order #1 (2/11/25) via MouserElectronics (\$53.95)</u> • PCM1794A • CCHD-957 • OPA1612 (1 more pending)			



ARCHIVE



PRELIMINARY SCHEMATICS SUBMITTED FOR PROPOSAL

Preliminary Schematic Overview:

12 VL53LOCX proximity sensors connected to one ESP32.

PCM5102 DAC > ESP32 > PAM84030 Audio Driver > R/L External Speakers.

Note for modifications in the next version:

2 dedicated ESP32s for processing sensor data

VL53L4CX substitute

PRELIMINARY SCHEMATIC OVERVIEW

** power*

- 3.3V power supply
 - VCC + GND
- 20 power (VDD) terminals to 3.3V
 - 7 ESP32s, 11 sensors, 1 microphone, 1 OLED display
- External power for subwoofer & speaker

** inputs*

- *sensors*
 - VIN to 3.3V
 - GND to GND
 - SCL to ESP32 SCL
 - SDA to ESP32 SDA
 - GPIO (none)
 - XSHUT to unique GPIO on ESP32
- *microphone*

** processors*

- ESP32s
- amplifiers
- DAC (Digital-to-Analog Converter)

** outputs*

- speakers
- subwoofer
- oled display

REFERENCES

A quick development platform for ESP32-A1S digital effect pedal...!

[repository](<https://github.com/hamuro80/blackstomp>)

- *Taptempo Stereo (Ping-Pong) Delay*
- *Stereo Chorus*
- *BLE Pedal*
- *MIDI Pedal*
- *Mic Mixer*
- *Gain Doubler*
- *Distortion*

An open source, Arduino compatible, ESP32-based Sample Player and Audio Framework.

[repository](<https://github.com/marchingband/wvr?tab=readme-ov-file>)

microphone and audio processing

[library](<https://github.com/sheavey/ESP32-AudioInI2S>)

ALLOCATION

ESP32 Processor #1: Data collection from ToF sensors

Processor #2: Data collection (...)

#3: Waveform synthesizer & basic manipulations

#4: Undecided waveform manipulation

#5: Undecided waveform manipulation

#6: Oscillator generation for OLED display

#7 Audio output processor & mixer for speakers